

Recursion

5/5/25

Administrivia

- HW 7 (maze search and trees) due Wednesday night

Recursive methods

- Check for base case
- Otherwise, break the problem into 1 or more smaller problem instances. Solve them by calling the same method and use the results to solve the original problem.

Sample problem: Identify subsets
of the first n integers

Recursive list methods

- Instead of LinkedList doing all the work, it passes the work to the nodes
- They call each other recursively and do a small amount of work each
- The method on a Node returns the result for that Node and its successors

Where is the base case for the size operation?

- A. In RecursiveList's size method
- B. In Node's size method
- C. In EmptyNode's size method
- D. In NonEmptyNode's size method
- E. Not exactly one of the above

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Which of the following correctly implements contains in EmptyNode?

- A. return true;
- B. return false;
- C. return value.equals(val);
- D. /* EMPTY METHOD BODY */
- E. Not exactly one of the above

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D. / * EMPTY METHOD BODY */

E. Not exactly one of the above

Which of the following correctly implements contains in NonEmptyNode?

- A. `return true;`
- B. `return false;`
- C. `return next.contains(val);`
- D. `return value.equals(val);`
- E. Not exactly one of the above

Which of the following correctly implements contains in NonEmptyNode?

A. return true;

B. return false;

C. return next.contains(val);

D. return value.equals(val);

E. Not exactly one of the above (combo of C&D)